# High-Resolution Tropical Cyclone Rainfall Detection From C-Band SAR Imagery With Deep Learning

Shanshan Mu, *Member, IEEE*, Xiaofeng Li, *Fellow, IEEE*, Haoyu Wang, *Member, IEEE*, Gang Zheng, *Senior Member, IEEE*, William Perrie, and Chong Wang, *Member, IEEE*

***Abstract*— This article introduces an innovative deep-learning approach for retrieving tropical cyclone (TC) rainfall information from C-band Sentinel-1 synthetic aperture radar (SAR) imagery. We collected 17 SAR images under TC conditions from 2016 to 2021 and matched them with synchronous observational Next Generation Weather Radar (NEXRAD) Level-III data, forming a dataset of 302 689 data pairs for model development. The model inputs include SAR-measured physical parameters in normalized radar cross section (NRCS), texture features represented by the gray-level co-occurrence matrix (GLCM), and statistical parameters of VV-polarized NRCS. A deep-learning-based TC rain rate retrieval (TC3R) model, combining a convolutional network and a fully connected (FC) network, was developed to retrieve quantitative TC rainfall information effectively. The test results demonstrate that the TC3R model can offer reasonable and stable quantitative rainfall estimation, particularly effectively detecting areas with medium-to-heavy rainfall events (2.5–40 mm/h) in SAR images where the NRCS is significantly affected by rain. Furthermore, to offer valuable insights into the performance of the TC3R model, we analyzed results across TC events of different intensities as case studies. Our results show high structural similarity (SSIM) in rainfall patterns between SAR and NEXRAD across all cases, consistently achieving SSIM values above 0.67. Moreover, in areas where SAR signals are notably affected by rainfall, the SSIM index even exceeds 0.80. Finally, our model's performance was evaluated by comparing its results with the independent global precipitation measurement (GPM) data, demonstrating effective rainfall prediction, particularly for the primary spiral rain band, in the two cases analyzed.**

***Index Terms*— Deep learning, rainfall, synthetic aperture radar (SAR) imagery, tropical cyclone (TC).**

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Shanshan Mu and Haoyu Wang are with the Key Laboratory of Ocean Observation and Forecasting and the Key Laboratory of Ocean Circulation and Waves, Institute of Oceanography, Chinese Academy of Sciences, Qingdao 266071, China, and also with the College of Marine Sciences, University of Chinese Academy of Sciences, Beijing 100049, China (e-mail: mushanshan@qdio.ac.cn; wanghy@qdio.ac.cn).

Xiaofeng Li and Chong Wang are with the Key Laboratory of Ocean Observation and Forecasting, the Key Laboratory of Ocean Circulation and Waves, Institute of Oceanography, and the Center for Ocean Mega-Science, Chinese Academy of Sciences, Qingdao 266071, China (e-mail: xiaofeng.li@ieee.org; wangchong1@qdio.ac.cn).

Gang Zheng is with the State Key Laboratory of Satellite Ocean Environment Dynamics, Second Institute of Oceanography, State Oceanic Administration, Hangzhou 310012, China (e-mail: zhenggang@sio.org.cn).

William Perrie is with the Fisheries and Oceans Canada, Bedford Institute of Oceanography, Dartmouth, NS B2Y 4A2, Canada (e-mail: william.perrie@dfo-mpo.gc.ca).

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## I. Introduction

TROPICAL cyclones (TCs) are strong natural hazards associated with heavy rain and high winds, which can cause substantial damages, such as severe flooding, mudslides, and tornadoes in coastal regions [1]. Quantitative estimating of TC rain rates is essential for determining TC intensity, evaluating the dynamic process of air–sea interaction, and mitigating economic and human losses in coastal areas [2], [3].

Rapid advancements in satellite remote sensing technology are increasingly becoming a powerful means for TC monitoring [4], [5], [6], [7]. Numerous studies have utilized satellite observations, including scatterometers [8], [9], [10], [11], [12], synthetic aperture radar (SAR) [13], [14], [15], [16], altimeters [17], and spaceborne radiometers [18], [19], [20], [21], to conduct TC research. Ever since the first launch of SEASAT loaded with SAR in 1978, hurricanes have been extensively monitored [22], [23]. More importantly, SAR provides a uniquely high spatial resolution (sub-kilometer scales) for detailed and extensive hurricane observations. Furthermore, its swath (hundreds of kilometers) is typically sufficient to cover an entire TC [3].

SAR has been highly successful in studying ocean surface winds [15], waves [24], and ocean currents [25], among others [26]. However, quantitative research on rainfall information still faces many challenges. These SAR-related studies primarily focus on the normalized radar cross section (NRCS), a parameter that SAR receives from the sea surface roughed by various complex interwoven air–sea interface processes, including wind, wave, rain, and so on [3]. Accurate evaluation of the rainfall information is more helpful and essential for the wind and wave retrieval from C-band SAR imagery, particularly in TC conditions, as the strong wind area near the TC center is often accompanied by rainstorms [2], [13].

Many papers have initially focused on investigating the mechanisms of rain imaging on SAR rather than quantitatively inverting rainfall information. Rainfall impacts NRCS primarily through two aspects: atmospheric effects (volumetric

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scattering and attenuation) and changes in sea surface roughness [16], [27], [28]. Previous studies have tried to investigate the magnitude of these impacts by simplifying the underlying physical mechanisms [29], [30]. Generally, atmospheric attenuation by rain has a minimal impact on C-band radar backscattering at low-to-moderate rainfall but still has a marked effect at intense rainfall (>50 mm/h) [31]. In contrast, rain-induced sea surface roughness is significant even at low and moderate rain rates [31], [32], [33]. Previous studies have revealed a positive trend between C-band SAR backscattering and rainfall under light/moderate rain conditions, while the trend reverses for heavy rain [34], [35].

Besides the influence of rainfall intensity on scattering mechanisms, the relationship between SAR NRCS and rainfall also depends on variations in radar frequency, incidence angle, polarization mode, and local sea conditions [16]. Identifying rain events for C-band SAR is more challenging since it operates in a transition region where raindrops can either enhance or diminish backscattered radar power compared to L- or X-band radar [16]. In addition, rain is more likely to have a significant impact on C-band measurements at higher incidence angles [33]. Rain cells dampen SAR backscattering at low-to-moderate incidence angles and enhance it at high incidence angles.

Overall, those satellite oceanographic studies have confirmed the potential of utilizing SAR-derived information for global-scale research on rainfall intensity and patterns [32], [33], [34], [35]. However, due to the combined effects of various factors contributed by rainfall, it remains challenging to clarify the precise SAR imaging mechanism of the complicated air–sea interface through physical formulas. Consequently, quantitatively estimating TC rain rates from SAR imagery is inherently demanding [16].

In recent years, artificial intelligence has entered a stage of rapid development and has continuously demonstrated its potential in oceanographic research fields [36], [37], [38], [39], [40]. Deep-learning networks have recently been applied to rain regime classification [41]. Unlike traditional methods that simplify physical mechanisms, deep learning can establish the relationship between observational data and target parameters through deep neural networks. Moreover, deep-learning networks possess powerful learning and representation capabilities, making them highly effective for solving regression problems [36]. Therefore, given the complexity of the SAR rain imaging mechanism, we aim to conduct quantitative rainfall inversion research based on deep-learning technology.

The article is organized as follows. Section II introduces the dataset creation. Section III shows the SAR image processing method and the TC rain rate retrieval (TC3R) model development. Section IV shows the model results and discussion. and Section V gives a conclusion.

## II. Dataset

For this study, we collected three kinds of data to develop the rain rate retrieval model. Seventeen Sentinel-1 SAR images and collocated Next Generation Weather Radar (NEXRAD) measurements constituted the matched dataset used for model tuning. Global precipitation measurement (GPM) data were also used to validate the model's application further.

### A. Sentinel-1 SAR Images

Sentinel-1 is a constellation of Earth observation satellites developed by the European Space Agency (ESA) as part of the Copernicus program. The Level-1 Ground Range Detected Sentinel-1 SAR images acquired at interferometric-wide (IW) mode and extra-wide (EW) mode with dual-polarization (VV/VH) were used in this study. The pixel spacing and swath of an IW/EW image are 10/40 m and 250/400 km, respectively. And the IW mode (EW mode) covers incidence angles from 30° to 46° (18.9° to 47.0°). A total of 17 C-band (∼5.3 GHz) SAR images over a six-year span (2016–2021) were used in this study. Among these, only two were acquired in EW mode because ground-based NEXRAD data, used as truth data, predominantly cover coastal and land areas where IW is the main acquisition mode.

These images, captured at various stages of TC development, include five tropical storms, five Category 1, two Category 2, three Category 3, and two Category 4 hurricanes, based on the Saffir-Simpson hurricane wind scale at the time of acquisition. The varying intensities assist in developing a more comprehensive rainfall inversion model tailored for TCs at different development stages.

For example, Fig. 1 shows a Sentinel-1 SAR image acquired over Hurricane Sally on September 15, 2020. The image shows two clear patterns on the right side, induced by spiral rain bands around the TC eye. In addition, the rain band features appear as darker stripes on the near side of the TC eye and brighter stripes on the far side, contrasting with the surrounding areas in the SAR image.

### B. NEXRAD Observations

The NEXRAD ground-based radar network contains 160 high-resolution S-band (∼2.4 GHz) Doppler weather radars and 34 radar stations distributed in the coastal area of America [13], [42]. NEXRAD can detect precipitation and wind, providing different forms and levels of precipitation data and products, which can be accessed through the National Centers for Environmental Information of the National Oceanic and Atmospheric Administration. NEXRAD has over 75 Level-III (L3) products. Among them, seven are precipitation products containing 1-h precipitation, storm total precipitation, digital instantaneous precipitation rate (DPR), etc.

Our study relies on the NEXRAD L3 DPR products, which provide a hybrid scan polar grid of instantaneous dual-polarization quantitative precipitation estimation precipitation rates to the nearest 0.001 in/h, for rain rate truth data [42], [43]. The DPR product displays the radar station-centered instantaneous precipitation rates with the radar scan time, resolution, and maximum coverage of 5 min, 250 m range by 1° azimuth, and 230 km. Notably, the DPR product is assembled from the lowest elevation range bins that can be used to compute a precipitation rate. A map of NEXRAD L3 data acquired adjacent to the SAR observation time is shown

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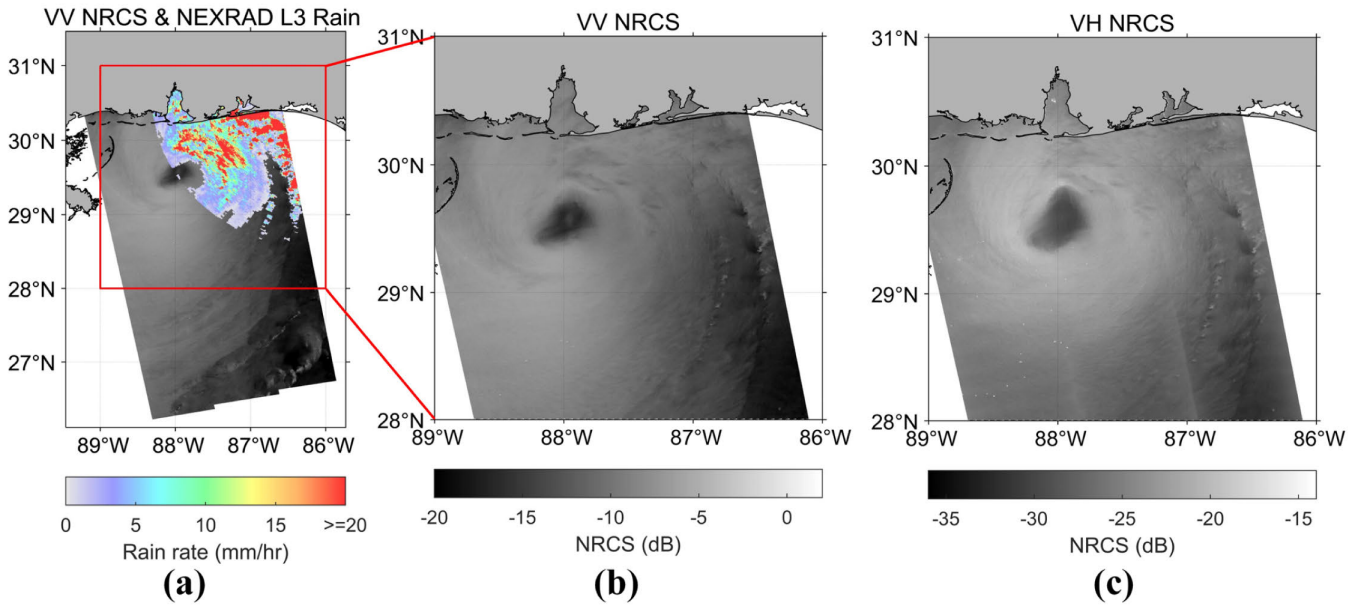



Fig. 1. Sentinel-1 SAR image acquired on September 15, 2020, over Hurricane Sally. (a) SAR NRCS in VV polarization overlaid with NEXRAD L3 rain rate product; zoomed on the red box area in (a) with SAR NRCS in (b) VV polarization and (c) VH polarization.

in Fig. 1(a). We can see that the spiral rain bands monitored by the two instruments have a good consistency.

### C. GPM Data

GPM is an international satellite mission that carries the Ku (∼13.6 GHz)/Ka (∼35.5 GHz) band dual-frequency precipitation radar and a multichannel GPM microwave imager to provide global observations of rain and snow. The integrated multisatellite retrievals for GPM (IMERG) is the unified algorithm that provides the multisatellite Level-3 gridded precipitation product [44]. IMERG effectively captures moderate to heavy rainfall but may struggle with light precipitation and very high-intensity events [44], [45].

Our case studies use GPM IMERG final precipitation Level-3 product as independent test data. It provides half-hourly precipitation estimates at $0.1^\circ \times 0.1^\circ$ and is systematically used to validate the model performance.

### D. Matched Datasets

In this article, we utilize the matching dataset of NEXRAD L3 and SAR for constructing the TC3R model, with independent validation conducted using GPM data. The matching process between NEXRAD L3 and SAR data is outlined as follows. First, we selected NEXRAD maps that closely matched the timing of SAR imaging, preferably those captured before SAR imaging, and the time difference for all matched data in this study is within 5 min. This occurs because rainfall moves at a certain speed, and while SAR captures sea surface rainfall information, NEXRAD detects rain cells at specific radar elevations. Consequently, when observations are made simultaneously, SAR will record signals after NEXRAD due to this timing difference. Finally, 17 SAR images were matched with NEXRAD L3 maps, resulting in 302 689 data pairs. Out of these, 56 668 data pairs from five SAR images under different TC conditions were used as test data, while the remaining data pairs were used for model training and validation. The data volume of the matched datasets in different rainfall intervals is shown in Fig. 2. Additionally, we conducted third-party independent validation by selecting one SAR image and its corresponding GPM data. This validation aimed to verify the effectiveness of the TC3R model further.

## III. Methodology

This section will detail the proposed method for deriving rain rates from SAR data, including data preprocessing and constructing the TC3R model.

### A. SAR Image Processing and Input Parameter Acquisition

Fig. 3 shows the overall flowchart of the dataset acquisition process. We first processed the SAR data using the ESA SNAP tool, which includes radiometric calibration, thermal noise removal, and land masking. In addition, a moving averaging window of $2 \times 2$ ($8 \times 8$) pixels was performed on EW (IW) data to remove the speckle noise and keep EW and IW data at the same spatial resolution of 80 m.

Then, to get the input parameters and better highlight kilometer-scale phenomena like TC rainfall [46], we split the SAR image into small vignettes through a moving window of $12 \times 12$ pixels and calculated the input parameters based on those SAR vignettes. The current step conducts downsampling on SAR images, changing the pixel size from $80 \times 80$ m to $960 \times 960$ m (12 pixels $\times$ 80 m/pixel).

As is known, the impact of rain on NRCS at the C-band is quite complex and depends on rain rate, sea state, incidence angle, and polarization [28], [47], [48], [49]. Fig. 1 shows that rain band features can be visually recognized based on image texture. Therefore, to ensure the comprehensiveness and effectiveness of the input parameters—since these are crucial

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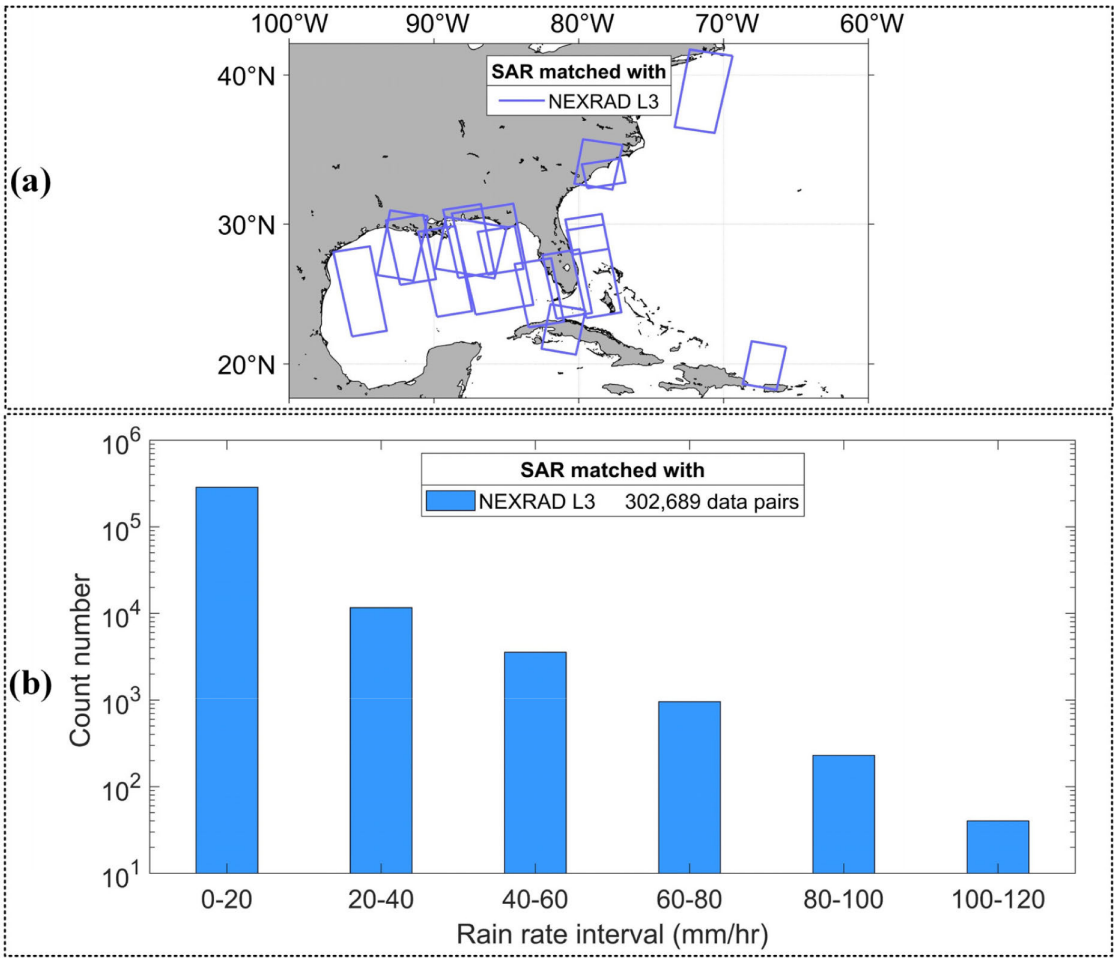



Fig. 2. Distribution and volume of the matchup datasets. (a) Locations of matched SAR data. (b) Histogram of rain rates derived from NEXRAD L3 data for the co-located dataset.

for the performance of deep-learning models—we considered physical, textural, and statistical features. Ultimately, this study selected seven parameters as inputs for the TC3R model.

The three basic physical parameters include the averaged SAR NRCS of VV and VH polarizations ($\sigma_0^{\mathrm{VV}}$, $\sigma_0^{\mathrm{VH}}$), as the combination of VV and VH polarizations has shown a higher efficacy in rainfall detection [31], along with the incident angle ($\theta$). The standard deviation of $\sigma_0^{\mathrm{VV}}$ ($\mathrm{STD}^{\mathrm{VV}}$) was incorporated as the statistical parameter. This parameter provides insights into image homogeneity and sea state. It is important to note that the statistical parameter and the following texture parameters were only extracted from SAR images in VV polarization due to its superior signal-to-noise ratio compared to VH polarization [13], [50], [51], [52], [53]. Despite thorough thermal noise removal in VH-polarization images, they still tend to be affected by noise during texture extraction. Expect that we add the important textural features calculated from the gray-level co-occurrence matrix (GLCM), which is a statistical method used to examine the texture of an image by considering the spatial relationship between pairs of pixels at specific distances and angles [54]. We select the three most commonly used GLCM-based texture feature parameters, Energy, Contrast, and Homogeneity, to enhance the quantitative extraction of rainfall information from SAR images.

The statistical and texture parameters are defined in the following. STD is a reflection of the degree of dispersion between pixel values and mean values of the SAR sub-image

$$\mathrm{STD} = \sqrt{\frac{1}{N-1}\sum_{i=1}^{N}|\sigma_{0i} - \overline{\sigma_0}|^2} \tag{1}$$

where $N$ is the number of pixels, $\sigma_{0i}$ and $\overline{\sigma_0}$ are each pixel value and mean value of SAR $\sigma_0$ vignette. Energy, Contrast, and Homogeneity reflect the following aspects: Energy returns the uniformity of image intensity distribution, Contrast measures the intensity contrast between a pixel and its neighbor and demonstrates the clarity of the image and the depth of the texture grooves, and Homogeneity returns a value that measures the closeness of the distribution of elements in the GLCM to the GLCM diagonal. They are all calculated based on the GLCM created from the sub-SAR image [54]

$$\mathrm{Energy} = \sum_{i=1}^{M}\sum_{j=1}^{M} P^2(i,j) \tag{2}$$

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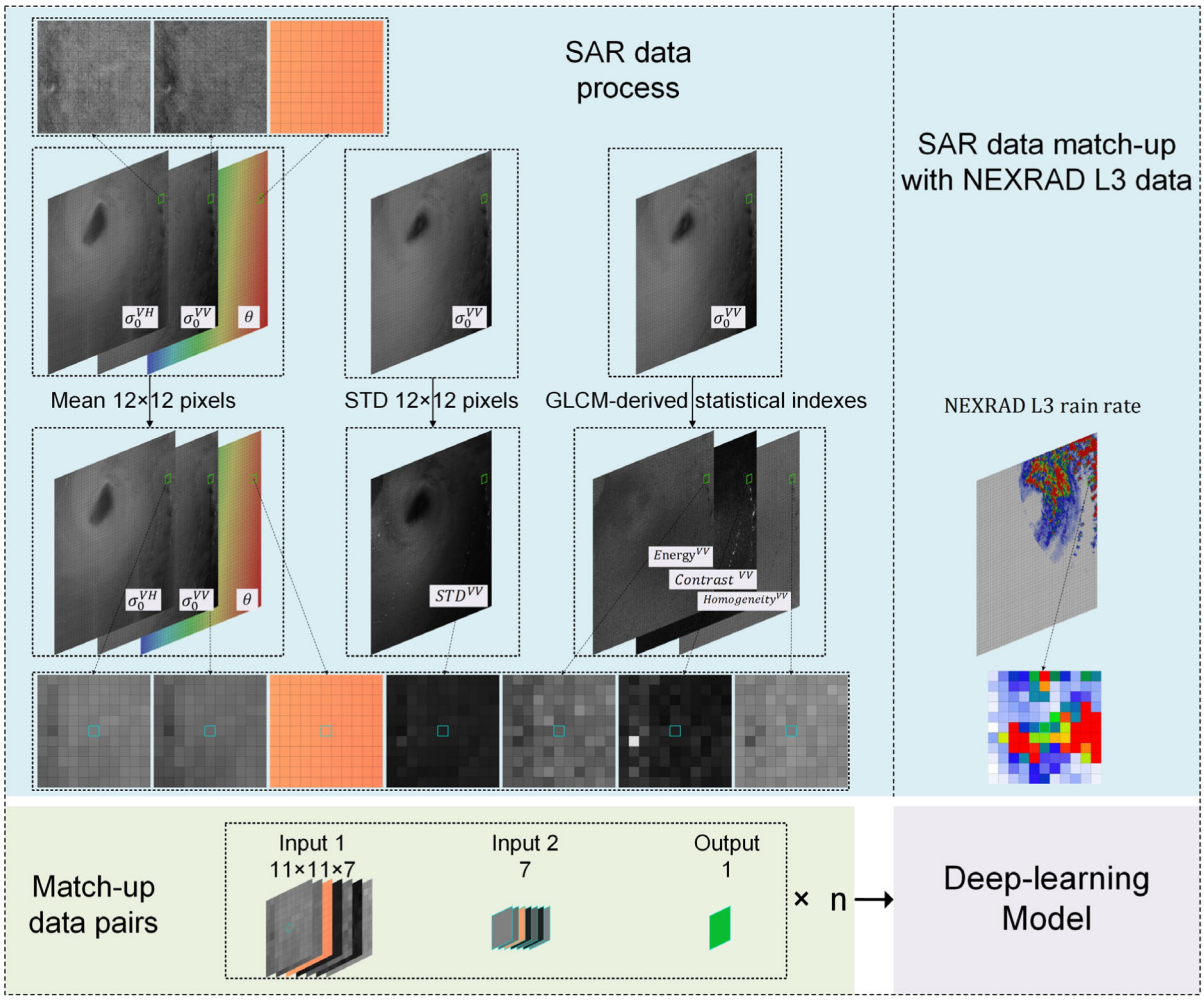



Fig. 3. SAR data processing and data pair creation flowchart. $\theta$ = incidence angle; $\sigma_0^{VV}$ and $\sigma_0^{VH}$ = NRCS with VV and VH polarizations; $\text{STD}^{VV}$ = standard deviation of $\sigma_0^{VV}$.

$$\text{Contrast} = \sum_{i=1}^{M}\sum_{j=1}^{M}(i-j)^2 P(i,j) \tag{3}$$

$$\text{Homogeneity} = \sum_{i=1}^{M}\sum_{j=1}^{M}\frac{P(i,j)}{1+|i-j|} \tag{4}$$

where $P(i, j)$ is the value of the pixel pairs $(i, j)$ in the GLCM.

### B. Model Development

Fig. 4 shows the framework of the deep-learning-based TC3R model. Our model constructs two input branches. Considering the time difference between NEXRAD and SAR imaging, which can lead to slight differences in rainfall location and the spatial continuity of rainfall information in SAR images, we designed the first input branch. This branch serves as the convolutional layer input, accepting 2-D image data of a sub-image region of 11 × 11 pixels (≈10 × 10 km) surrounding the target pixel. Given that the extent of TC rain bands is typically 10 km, we choose the five-pixel region surrounding the target to increase the model's receptive field. This branch comprises three convolutional layers, followed by a BatchNorm2d (BN) layer, a MaxPool2d (MP) layer, and an AdaptiveAvgPool2d (AAP) layer. The BN layer normalizes outputs for stable and faster training, the MP layer reduces spatial dimensions by selecting maximum values, and the AAP layer ensures a fixed-size output by averaging each region, providing consistent input for the fully connected (FC) layers.

The second input branch is the FC layer branch, which takes 1-D SAR variables as input. This branch is primarily designed to extract rainfall-related information corresponding to the input parameters of the target pixel. It does this through multiple hidden layers, including an FC layer and two basic blocks. The basic Block represents a residual mapping process designed based on the core concept, residual learning, of the Residual network [55]. The basic Block process is detailed in the bright pink box at the bottom. The original input passes through shortcut connections to the next two layers. This design primarily aims to reduce information loss during training and optimize model performance.

The outputs of the two branches are then concatenated and input to the subsequent part, represented by the green box in Fig. 4, consisting of two FC layers and two basic Blocks. The combination of the two branches effectively utilizes convolutional and FC networks to extract information from various directions, establishing connections with the output

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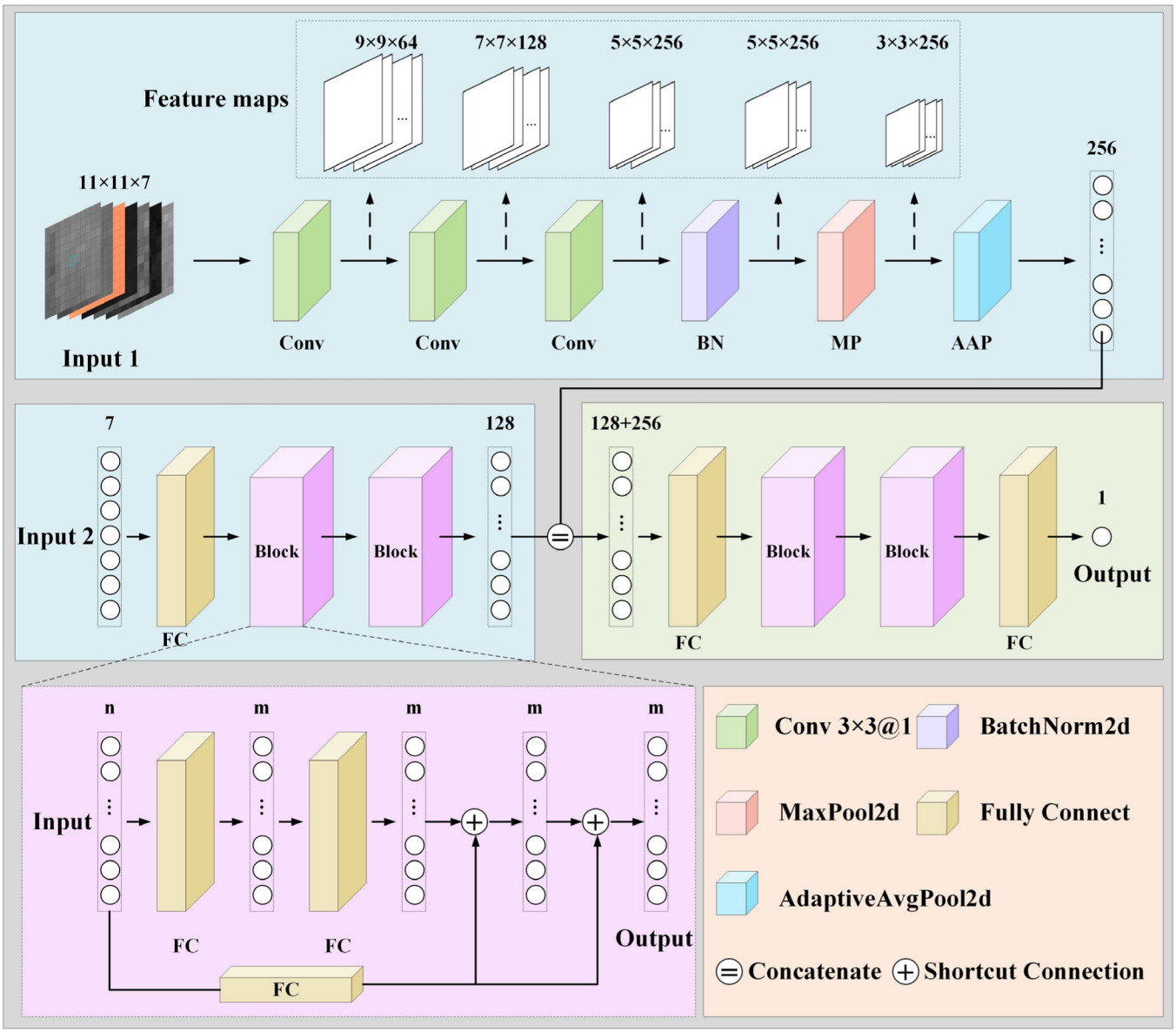



Fig. 4. Flowchart of the proposed deep-learning-based TC rainfall retrieval model. The numbers at the top indicate each layer's size and number of channels. For example, 11 × 11 × 7 denotes an input of 11 × 11 sub-images with seven channels; numbers in parentheses, like 7 and 128, represent the number of neurons in the current layers. Note that the activation function used in the model is ReLU, and ReLU activation is applied to each FC layer and convolutional (Conv) layer. And 3 × 3.1 indicates a convolutional kernel size of 3 × 3 with a stride of 1.

information to enhance the optimization performance of the model. Notably, the model employs ReLU, one of the most commonly used nonlinear activation functions, applied after each FC and convolutional layer in the entire TC3R model architecture.

The symbols in the TC3R model architecture in Fig. 4 are explained as follows. In the upper panel, the numbers at the top indicate each layer's size and number of channels. For example, 11 × 11 × 7 denotes an input of 11 × 11 sub-images with seven channels, and so forth. The convolutional layers have a kernel size of 3 × 3 with a stride of 1, represented by the symbol 3 × 3.1 in the green boxes. The numbers 7 and 128 in the central panel represent the number of neurons in the corresponding layer. For instance, 7 signifies the number of input neurons. The box at the bottom-left panel explains the internal structure of each basic block represented by the bright pink blocks. The bottom orange box provides explanations of the specific contents of the abbreviations.

We chose Rectified Adam as the optimization algorithm, which addresses some limitations of the Adam optimizer, particularly its variance and stability during early-stage training. The mean-squared error (mse) was selected as the loss function, measuring the average squared difference between targets and predictions. mse is commonly used in regression networks and is well-suited for the gradient-descent-based optimization algorithm employed in this study, aiding model convergence as the error and gradient decrease. Regarding the training strategy, we employed early stopping to maintain the model's generalization [56]. Early stopping is a regularization technique that halts training when the validation set performance stops improving, preventing overfitting.

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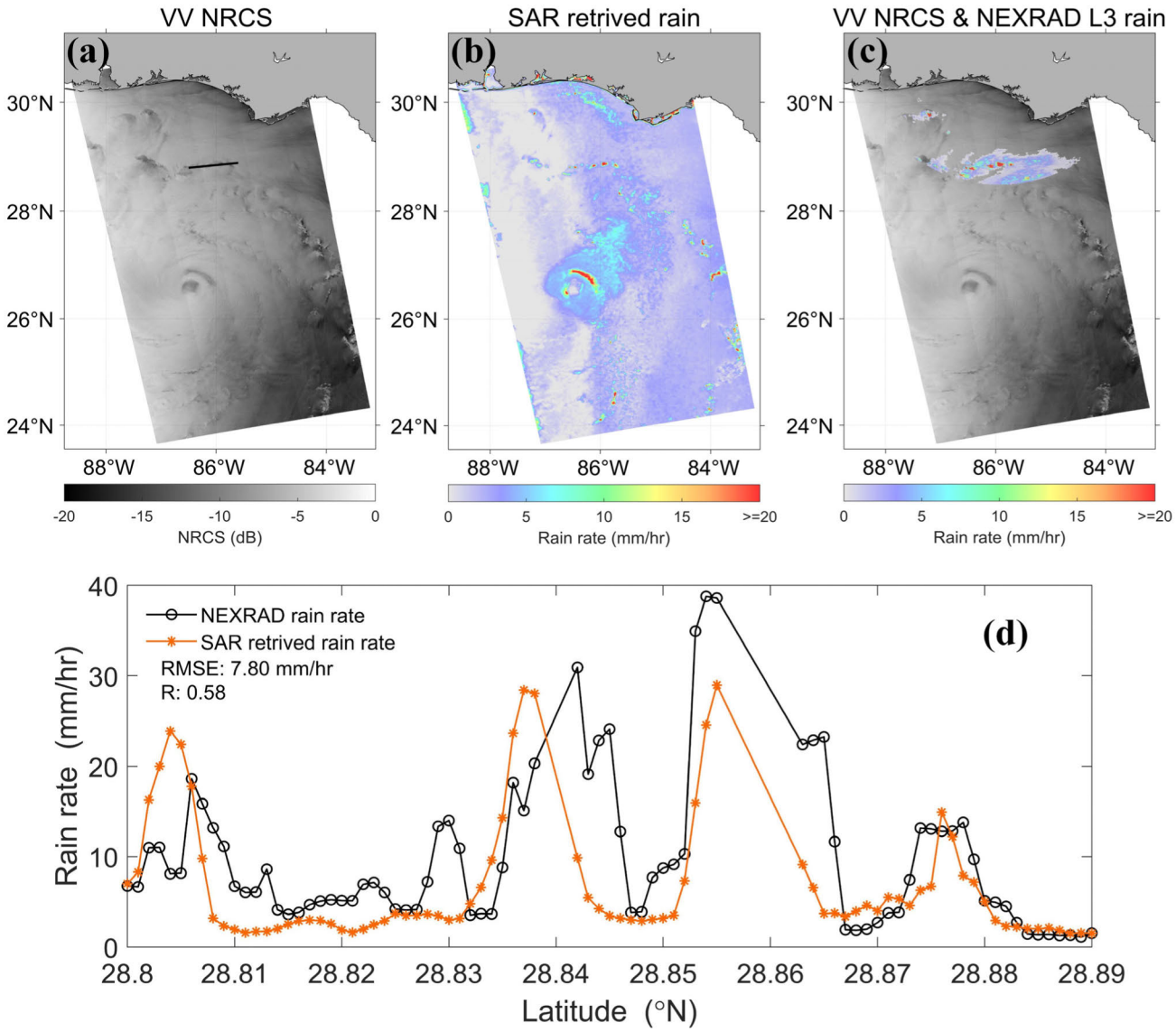



Fig. 5. Maps of SAR NRCS measured over Hurricane Michael on October 9, 2018, and the comparison results between SAR-retrieved rain and NEXRAD L3 data. (a) NRCS in VV polarization. (b) SAR-retrieved rain rate. (c) NEXRAD L3 rain rates. (d) Line chart comparing SAR and NEXRAD rain rates along the black cross-sectional line in (a).

## IV. Results and Applications

### A. Model Performance

Here, the deep-learning-based TC3R model was rigorously tested. Due to the time lag between NEXRAD and SAR imaging, minor discrepancies in rainfall locations may arise, making pixel-level comparisons challenging [30]. Therefore, in this section, we selected three of the five independently tested SAR datasets to showcase the spatial distribution of SAR rainfall and visually compare it with NEXRAD rainfall data. Additionally, we chose a cross-sectional profile spanning high, medium, and low rainfall areas for each case to conduct statistical analyses of both rainfall datasets within the profile.

The three SAR images captured during Hurricane Michael on October 9, 2018 (Fig. 5), Hurricane Irma on September 10, 2017 (Fig. 6), and Hurricane Elsa on July 6, 2021 (Fig. 7) serve as validation datasets.

Precipitation manifests in various regions within TCs. The strongest precipitation distribution typically occurs around the eyewall because this region experiences the most intense convection. Fig. 5(a) shows a prominent northwest-southeast black band around the eyewall at 26.5°N. This feature corresponds to localized heavy rainfall areas, affecting the intensity of the NRCS signal compared to the surrounding areas. As can be seen from Figs. 5(a)–7(a), precipitation also concentrates in spiral rain bands extending outward from the TC center; the signal of the irregular shapes of meandering rain bands is distinguishable in SAR VV-polarized NRCS images. These rain bands present a spiral strip shape and have higher or lower backscatter signals than the surrounding pixels.

Figs. 5(b)–7(b) present the rainfall maps retrieved by our TC3R model. A visual comparison reveals a close match between the estimated rainfall intensity spatial patterns and the radar backscattering signal shown as the bright feature

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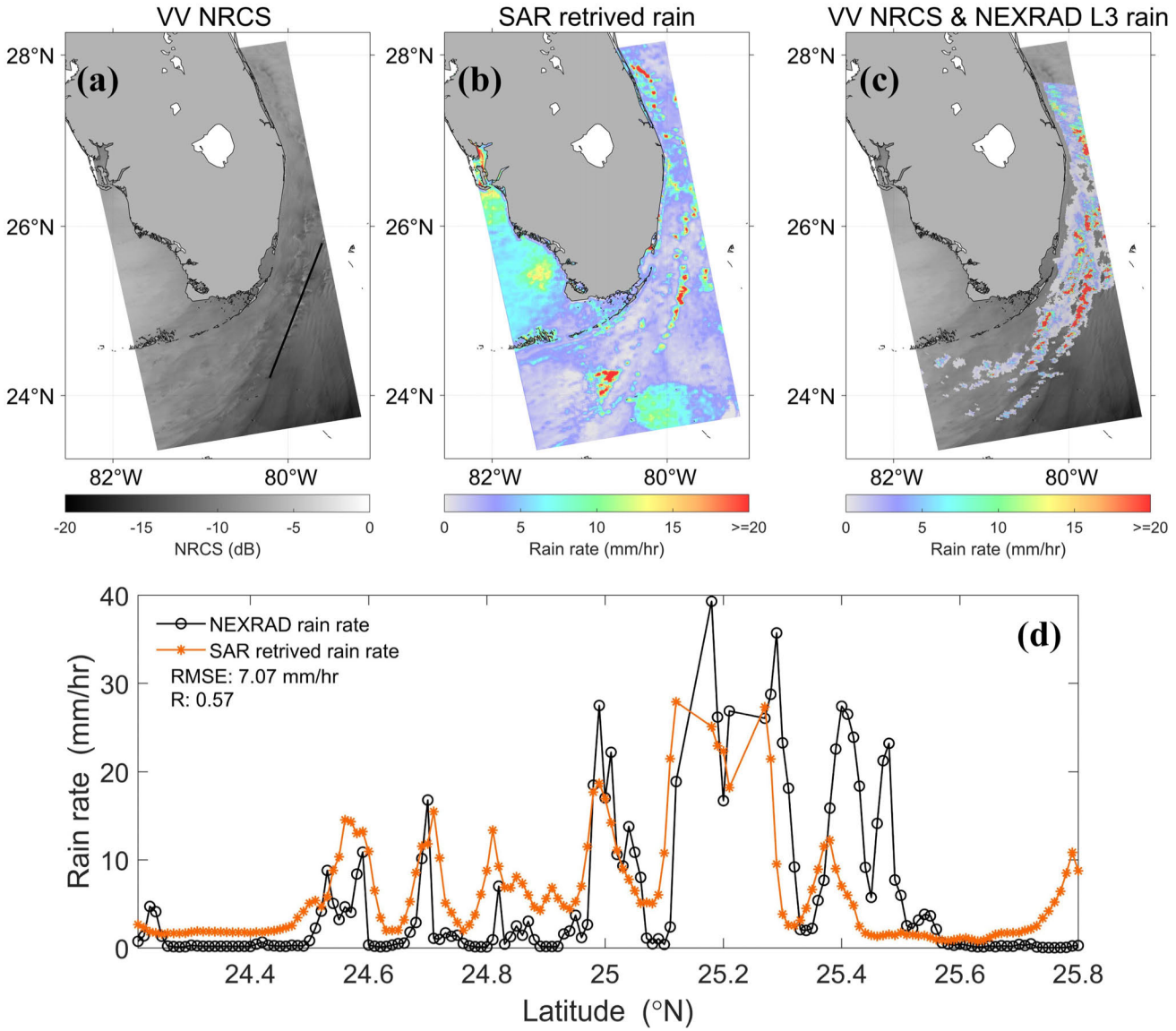



Fig. 6. Maps of SAR NRCS measured over Hurricane Irma on September 10, 2017, and the comparison results between SAR-retrieved rain and NEXRAD L3 data. (a) NRCS in VV polarization. (b) SAR-retrieved rain rate. (c) NEXRAD L3 rain rates. (d) Line chart comparing SAR and NEXRAD rain rates along the black cross-sectional line in (a).

and the distinct black narrow bands around the TC eye visible in Figs. 5(a)–7(a). Figs. 5(c)–7(c) show the corresponding NEXRAD L3 rain rate product superimposed onto the SAR images in VV polarization. As shown, the SAR-retrieved rainfall information matches well with the rainfall intensity spatial distribution provided by NEXRAD, especially for the backscattered signals on SAR images affected by rainfall. This alignment indicates that our TC3R model effectively detects rainfall areas in SAR images, demonstrating the model's ability to discriminate between rain and non-rain regions in all three cases.

In Figs. 5–7, the RMSE results for the 2-D-matched data points between SAR and NEXRAD are 5.08, 9.00, and 11.34 mm/h, respectively. To further quantitatively validate the TC rain rates retrieved from TC3R and better observe its effectiveness in different rainfall intensities, we analyzed a cross-sectional line spanning high, medium, and low rainfall regions [see the black line in Figs. 5(a)–7(a)]. Figs. 5(d)–7(d) present the statistical results and comparison line chart along the selected transect of the rain band for three cases. Overall, the analysis demonstrates that our model provides dependable quality in rainfall inversion over low to high rainfall (rates below 50 mm/h) and maintains high consistency in the trend depicted in the line graph compared to NEXRAD data with RMSE and $R$ of 7.80, 7.07, 10.33 mm/h, and 0.58, 0.57, 0.63.

However, it is worth noting that our model, like any other, has its limitations. As shown in Figs. 5 and 6, the TC3R model demonstrates significant agreement with NEXRAD data, particularly when the rain rate is less than 30 mm/h. However, it struggles to capture rare heavy-to-violent rain events (rates $>30$ mm/h), and there is a tendency for underestimation in such rare areas of heavy rainfall. This is primarily due to the scarcity of samples for extreme rainfall events, which

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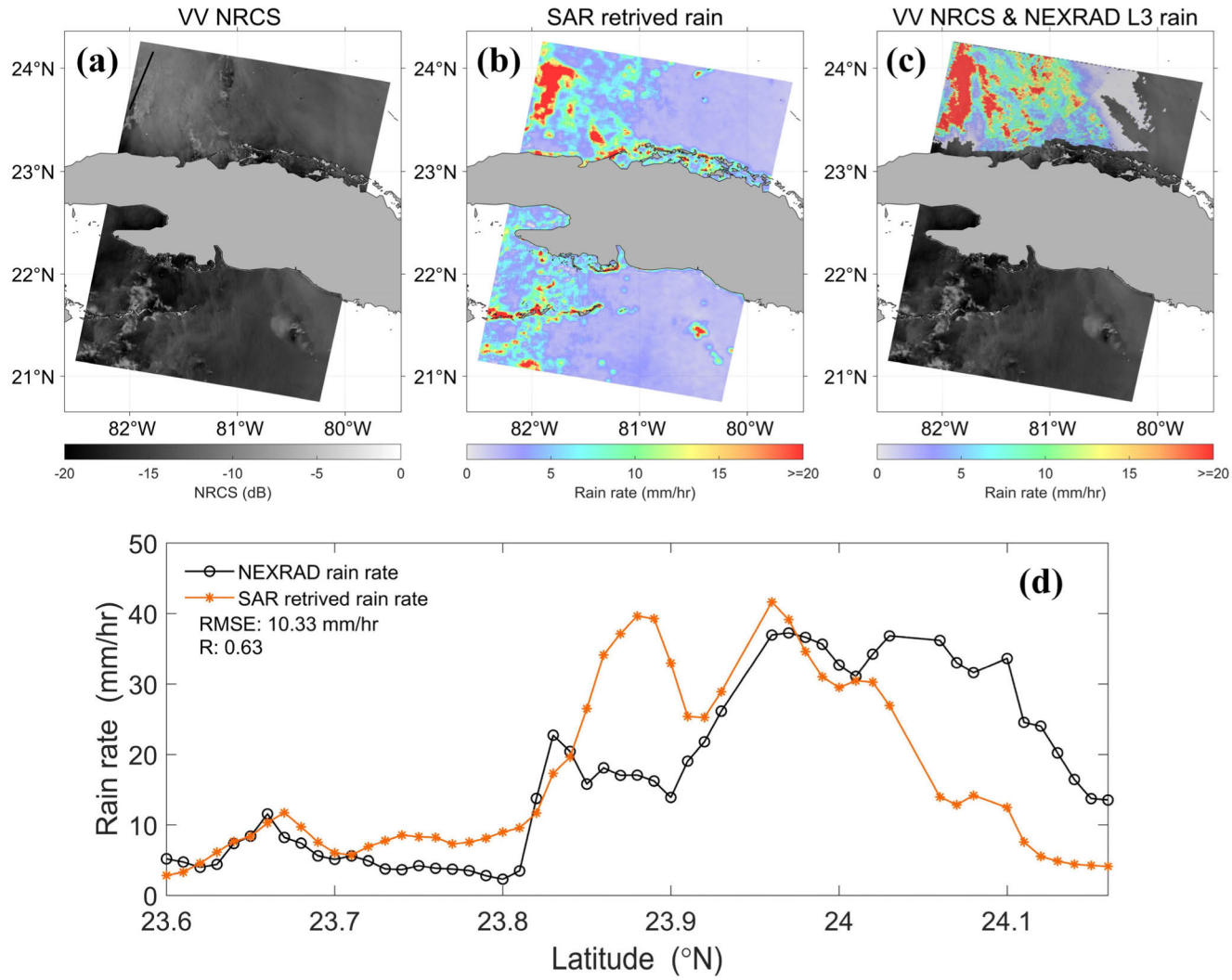



Fig. 7. Maps of SAR NRCS measured over Hurricane Elsa on July 6, 2021, and the comparison results between SAR-retrieved rain and NEXRAD L3 data. (a) NRCS in VV polarization. (b) SAR-retrieved rain rate. (c) NEXRAD L3 rain rates. (d) Line chart comparing SAR and NEXRAD rain rates along the black cross-sectional line in (a).

comprise less than 3% of the dataset. While TCs contribute significantly to heavy rainfall data, rain events with such high rainfall rates are still extremely rare. Moreover, the application of SAR technology often requires substantial investment and resource support, making it highly unlikely for space-borne SARs to capture all TC images [16]. These factors limit the effectiveness of our model in retrieving intense rainfall events from small samples, as deep-learning-based models are known to require balanced samples and a sufficient quantity of data.

### *B. Application of the TC3R for Varying Intensities of TCs*

To further demonstrate the performance of TC3R under various TC conditions, we selected three cases from the five test images with different intensities based on the Saffir-Simpson Hurricane Wind Scale as case studies to assess the model's performance. The selected instances include one tropical storm image, one Category 1, and one Category 2 hurricane image. Furthermore, we categorized instantaneous rainfall intensity into four levels based on the manual of surface weather observations (MANOBS) [57]: Light rain—rainfall less than 2.5 mm/h; moderate rain—rainfall rate ranging from 2.6 to 7.5 mm/h; heavy rain—rainfall rate exceeding 7.6–50 mm/h; violent rain—precipitation rates exceeding 50 mm/h. The classified results into four levels of rainfall intensity are depicted in Figs. 8–10. It is important to note that the ground-truth data does not fully cover the SAR image, resulting in some blank spaces due to the NEXRAD observation centered around the station. We only compared the regions where ground-truth data existed to compare the two more clearly.

In addition, we employed a statistical parameter, structural similarity (SSIM) [58], to better assess the similarity between the TC3R model output and NEXRAD rainfall. SSIM compares local patterns of pixel intensity and mainly considers three key characteristics of two images: luminance, contrast, and structure. It provides a more comprehensive indication of the similarity between the two images, ranging from 0 to 1 [58]. The larger the SSIM value, the more similarity between the two images. When SSIM = 1 indicates, they are the same. The calculation formula is

$$\mathrm{SSIM}(x, y) = \frac{(2\mu_x\mu_y + c_1)(2\sigma_{xy} + c_2)}{(\mu_x^2 + \mu_y^2 + c_1)(\sigma_x^2 + \sigma_y^2 + c_2)} \tag{5}$$

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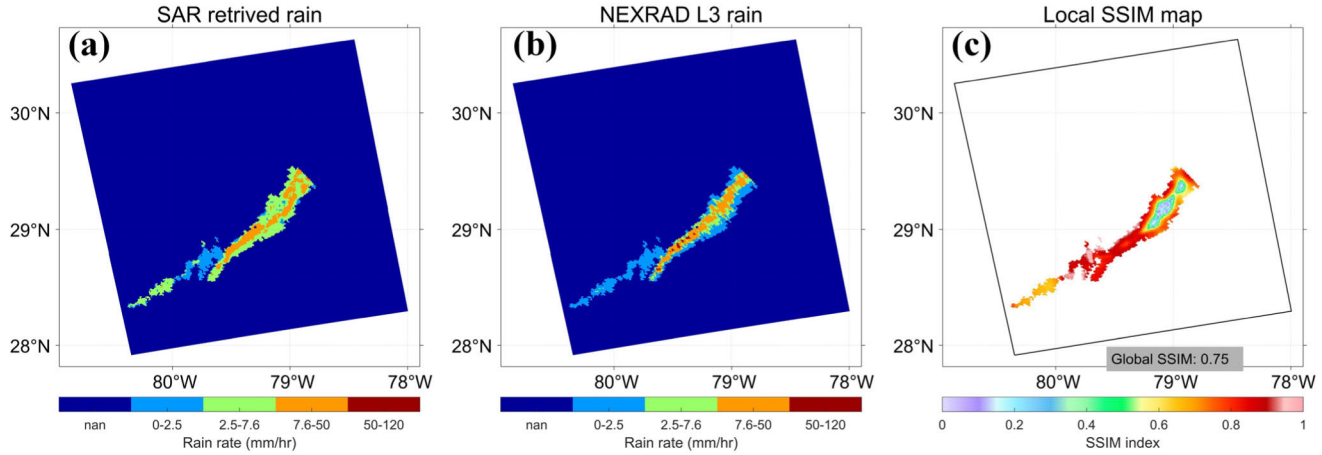



Fig. 8. Maps of (a) SAR-retrieved rain and (b) NEXRAD L3 rain as measured over Hurricane Eta on November 12, 2020, and (c) SSIM between (a) and (b).

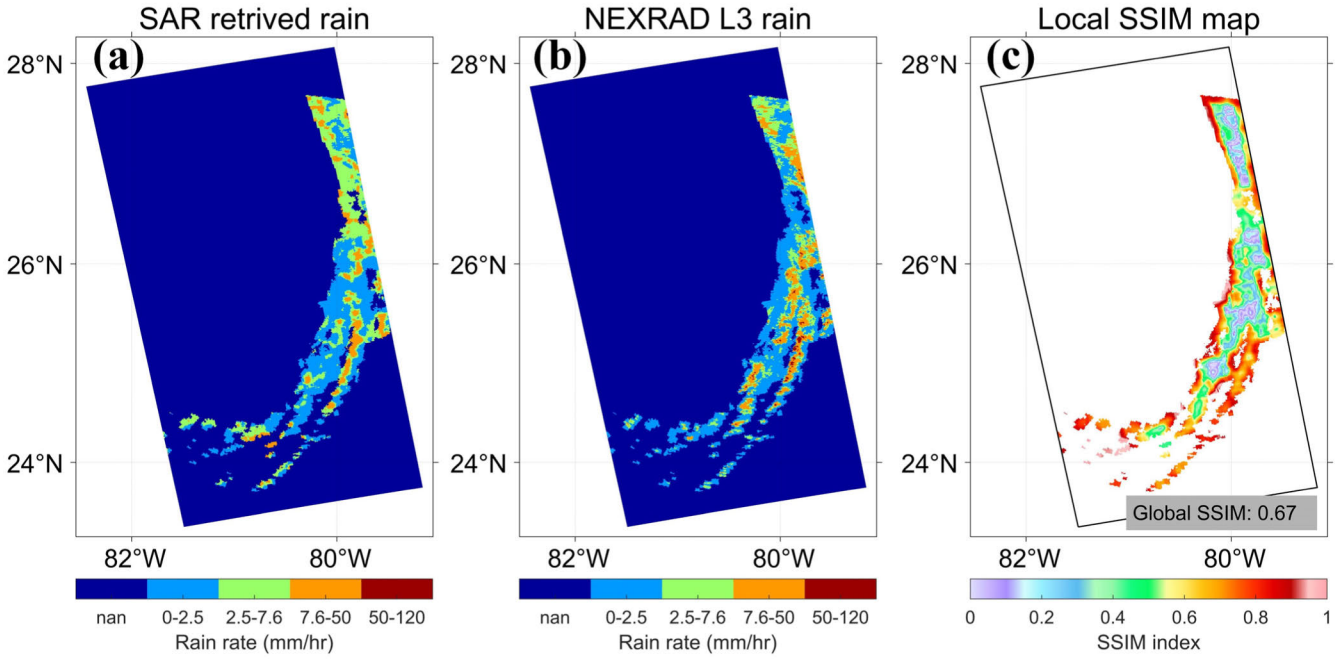



Fig. 9. Maps of (a) SAR-retrieved rain and (b) NEXRAD L3 rain as measured over Hurricane Irma on September 10, 2017, and (c) SSIM between (a) and (b).

$$c_1 = (0.01 * L)^2 \tag{6}$$

$$c_2 = (0.03 * L)^2 \tag{7}$$

where $\mu_x$ and $\mu_y$ are the mean values of the $N \times N$ $x$ and $y$ sub-image, respectively, $\sigma_x$ and $\sigma_y$ represent the $N \times N$ $x$ and $y$ sub-image variance, respectively. $\sigma_{xy}$ represents the covariance between the $x$ and $y$ sub-image. $L$ represents the dynamic range of the pixel values. This article scaled the rain rate image to 0–4, so $L$ equals 4.

We compare the rainfall maps retrieved by TC3R and provided by NEXRAD L3, categorized by rain rate, across three TC intensities. Fig. 8 shows a tropical storm over the Atlantic Ocean on November 12, 2020. Hurricane Eta was a powerful and devastating TC that impacted Central America and parts of the southeastern United States in November 2020. Fig. 9 illustrates the TC rainfall retrieved through SAR, alongside the corresponding NEXRAD rainfall conditions over Hurricane Irma, an extremely powerful hurricane, making landfall in South Florida. It was acquired during the dissipation stage of Hurricane Irma on September 10, 2017, and the acquisition intensity was Category 1. Fig. 10 depicts a SAR image acquired during Hurricane Ida before it landed in Louisiana, United States, which was classified as a Category 2 hurricane.

Overall, these comparisons, shown in Figs. 8–10, yield encouraging results. The global SSIM indices for the three cases are 0.75, 0.67, and 0.68. In keeping with [58], the relatively high value of SSIM indicates that the intensities and spatial structural patterns of the TC3R-retrieved rain align well with the NEXRAD-observed rain. The overall performance of the TC3R model shows that it can provide reasonable rainfall estimates for different hurricane intensities.

Comparing the rain fields in Figs. 8(a) and (b)–10(a) and (b), we find similar spatial rain patterns in the TC3R-retrieved and NEXRAD L3-provided rain. A comprehensive analysis of

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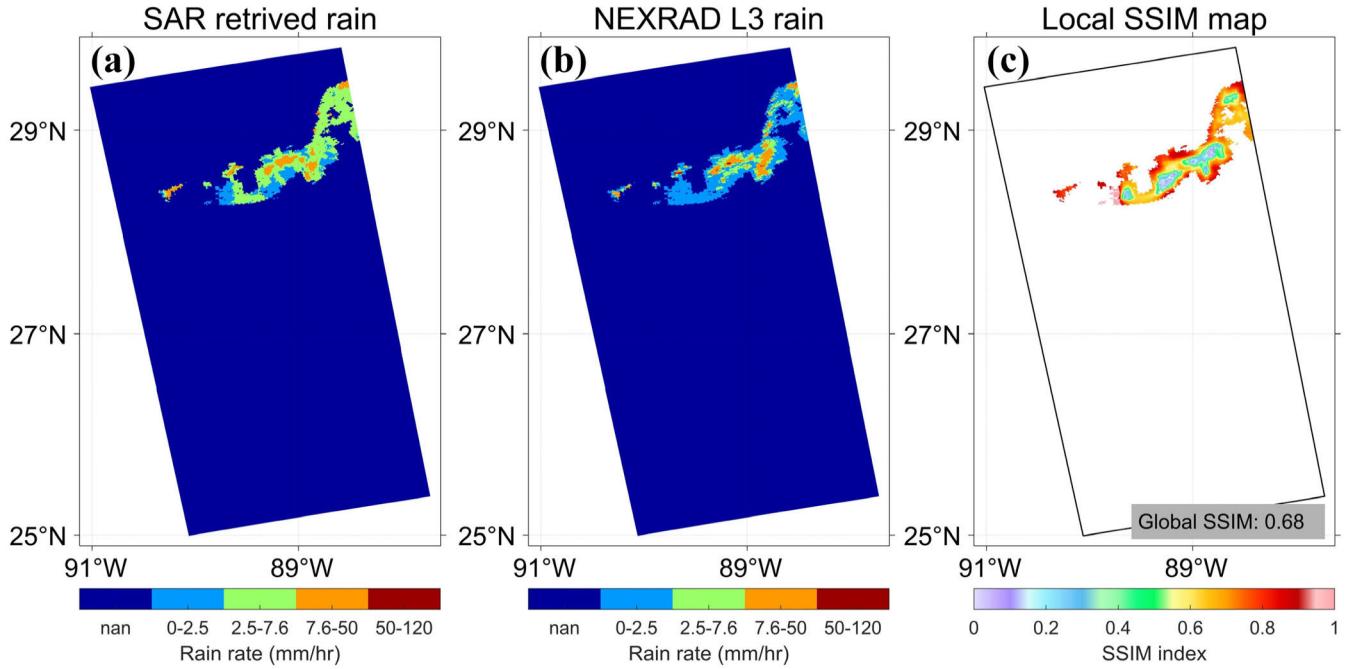



Fig. 10. Maps of (a) SAR-retrieved rain and (b) NEXRAD L3 rain as measured over Hurricane Ida on August 29, 2021, and (c) SSIM between (a) and (b).

TABLE I

MODEL PERFORMANCE FOR DIFFERENT RAINFALL INTERVALS BASED ON MATCHED DATA IN FIGS. 8–10

| Rainfall intervals | Matched data in Fig. 8 | | Matched data in Fig. 9 | | Matched data in Fig. 10 | |
|---|---|---|---|---|---|---|
| | Precision | N | Precision | N | Precision | N |
| **0 mm/hr < Rain rate ≤ 2.5 mm/hr** | 0.34 | 1735 | 0.58 | 10229 | 0.28 | 3858 |
| **2.5 mm/hr < Rain rate ≤ 7.6 mm/hr** | 0.58 | 510 | 0.45 | 3697 | 0.71 | 1135 |
| **7.6 mm/hr < Rain rate ≤ 50 mm/hr** | 0.54 | 751 | 0.28 | 2738 | 0.50 | 718 |
| **Rain rate > 50 mm/hr** | 0 | 38 | 0 | 182 | 0 | 27 |

*Note*. N = The number of truth data samples.

the three cases shows that SAR measurements indicate high-quality rain rates up to 50 mm/h in agreement with NEXRAD L3 products. The TC3R model performs best for moderate to heavy rain (2.5–50 mm/h), with local SSIM results above 0.60, even exceeding 0.80. Despite the different tropical storm ratings, all three cases exhibit violent rainfall, as shown in the NEXRAD rainfall map.

To better illustrate the performance of the TC3R model across different rainfall intervals, we provide the sample sizes and statistical results for different rainfall intervals based on the matched data shown in Figs. 8–10. The results are presented in Table I, indicating that the TC3R model is most effective at detecting moderate to heavy rainfall, consistent with the C-band SAR's sensitivity to rainfall signals. Precision, defined as the ratio of true positive (TP) predictions to the total number of TP and false positive (FP) predictions, measures the classifier's accuracy when predicting positive classes. It is calculated as

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{8}$$

where TP is the number of instances that are actually positive and were correctly predicted as positive, and FP is the number of instances that are actually negative but were incorrectly predicted as positive by the classifier.

However, as shown in Table I and Figs. 8–10, the comparison results show that the model performs poorest in detecting violent rainfall, whereas the TC3R model largely underestimates rain rates exceeding 50 mm/h. This is largely attributed to the limited number of extreme rainfall samples comprising less than 2%. Additionally, for low rainfall signals below 2.5 mm/h, the SSIM index for some localized light rainfall areas is relatively low, around 0.4. This result shows that our approach can better detect higher rain rates, where the SAR signal is strongly affected by rain than low to medium rain rates. This can be attributed to the inherent limitations of C-band SAR instruments, which have much lower sensitivity to light rain than X-band radar, as stated in [31].

### *C. Comparison of Rainfall Retrieved From SAR Images and GPM Rainfall*

To further validate the performance of the TC3R model and to select independent GPM data for qualitative assessment and comparative analysis of rainfall information obtained from SAR, we compare SAR-retrieved rain rates with collocated satellite-based GPM L3 half-hourly precipitation data closest to the SAR imaging time. Fig. 10 shows the comparison results of the two cases: Hurricane Dorian [Fig. 11(a)–(c)] and

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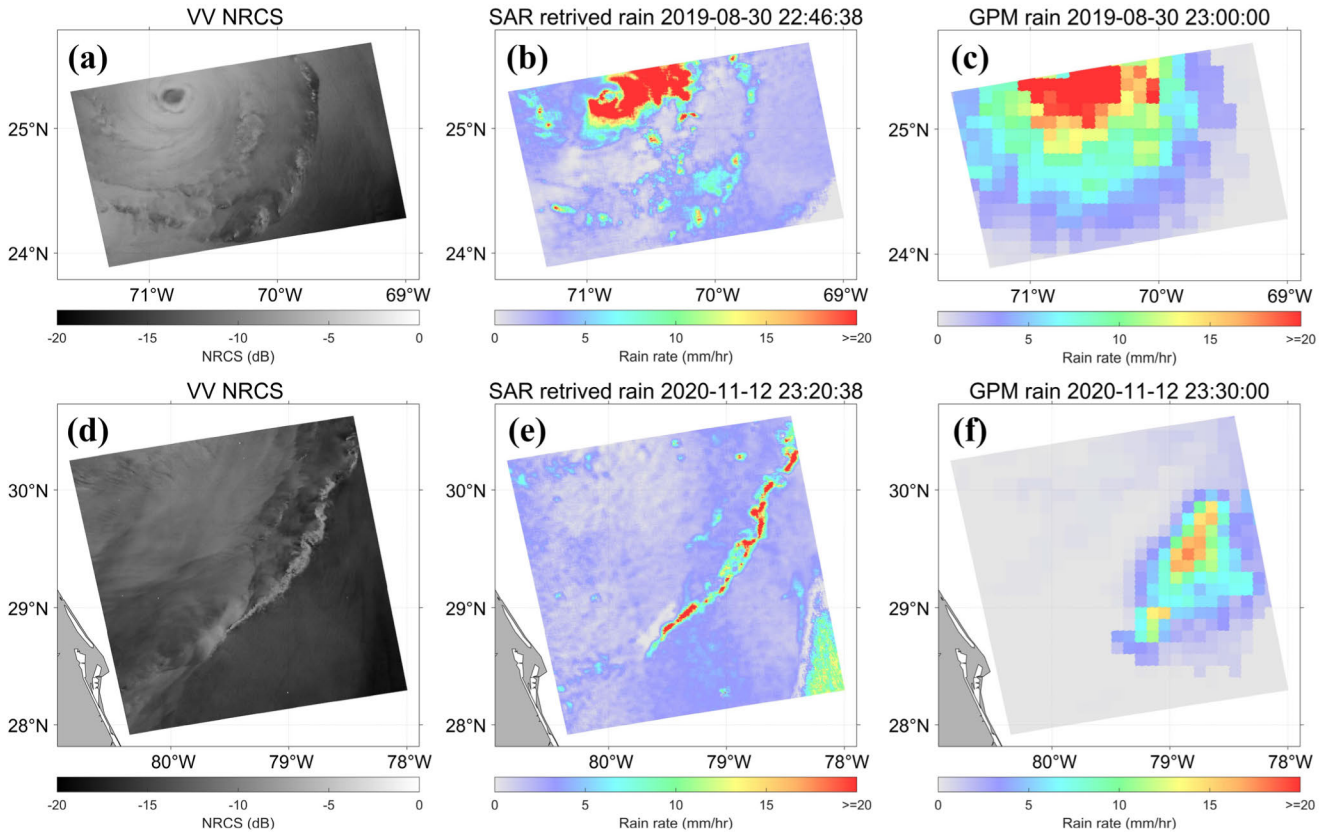



Fig. 11. Maps of SAR NRCS and rain rates over (a)–(c) Hurricane Dorian on August 30, 2019 and (d)–(f) Hurricane Eta on November 12, 2021. SAR NRCS of (a) Hurricane Dorian and (d) Hurricane Eta; rain rates derived from the TC3R model over (b) Hurricane Dorian and (e) Hurricane Eta; rainfall information provided by the GPM data over (c) Hurricane Dorian and (f) Hurricane Eta.

Hurricane Eta [Fig. 11(d)–(f)]. Overall, the spatial rain band patterns retrieved by the TC3R model [see Fig. 11(b) and (e)] align more closely with the squall line structures observed in the SAR NRCS images [see Fig. 11(a) and (d)]. On the other hand, the SAR-acquired rainfall rates are visually identified as high SSIM with the products provided by GPM [see in Fig. 11(c) and (f)] despite the resolution difference between the two datasets.

As shown in Fig. 11(a) and (b), the strongest rainfall surrounds the east side of the eyewall of Hurricane Dorian, as this is where the convection is strongest, followed by the outer spiral rain bands that wrap outward from the TC center. As can be seen, there is generally no precipitation in the clear TC center because it is a sinking area. The TC eye usually exhibits the lowest rainfall and wind speeds [3]. Comparing Fig. 11(b) and (c), we can observe that the TC3R model effectively captured the intense rainfall near the TC center, corroborating with the high rainfall areas identified by GPM.

Additionally, the inversion results for Hurricane Eta [Fig. 11(e)] show that the SAR images exhibit nearly identical trends and positions of rainfall structures as those provided by GPM [Fig. 11(f)], indicating that the TC3R model effectively captures signals heavily influenced by rainfall in SAR images. However, the two have discernible differences regarding coverage area and intensity. The rainfall bands retrieved from SAR inversion closely align with the visible and narrow rainfall regions provided by VV-polarized NRCS, offering rainfall location and intensity information at a resolution of nearly 1 km. In contrast, GPM provides $0.1^\circ \times 0.1^\circ$ rainfall information, resulting in considerable underestimation of rainfall intensity. Extremely heavy rainfall in localized areas is inaccurately observed under coarse-resolution instruments. Additionally, there is a certain observational time difference between the two; SAR provides instantaneous rainfall intensity information, whereas GPM provides half-hourly precipitation estimates, inevitably resulting in discrepancies in intensity measurements within this interval.

However, Fig. 11 shows obvious estimation errors on the right edge of the SAR image. One possible reason is that when the NRCS at cross-polarization falls below the noise floor of Sentinel-1A SAR, VH-polarization information may not provide reliable rainfall signals. The TC3R model considers the echo signal of VH polarization, and the right sub-swath of the VH signal is contaminated with noise. In contrast, the model is subjected to the noise signal during the convolution process and focused on the image edge and textural noise features, resulting in an incorrect estimation of rainfall information [48], [59]. This case also illustrates that our method does not only detect areas impacted by rain but also by non-wind-related features that produce inhomogeneity in the SAR image. Indeed, areas with low backscattered values associated with low winds, such as in the island's lee (middle) or close to the coast (south part of the area), are detected by the TC3R model because the detection focuses on edges and textural features.

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### D. Discussion

This study contributes to retrieving rain rates under TC conditions through the TC3R model by designing improved input parameters and model architecture. The TC3R model considers the basic SAR physical information such as $\sigma_0^{VV}$, $\sigma_0^{VH}$, $\theta$ as well as the statistical information of SAR echo signals $STD^{VV}$ and the texture information of SAR backscatter signals: Energy, Contrast, and Homogeneity. Additionally, the model combines data from the target pixel and a surrounding $11 \times 11$-pixel receptive field, effectively capturing the scale of hurricane rain bands. However, this strategy limits the model's applicability over land and coastal regions due to interference from land signals, making rainfall monitoring within approximately 5 km of land challenging.

The research results demonstrate the improved capability of deep learning in high rainfall conditions. However, there is still room for improvement in inverting extremely rare and intense rainfall, which depends on the number of rare rainfall samples (>30 mm/h) that account for only 2.81% of the data in this study. In addition, the TC3R model performs better in detecting visually identifiable spiral rain bands, where the impact of rainfall signals is more pronounced, compared to extracting strong rainfall areas near the eyewall. The limitations of the TC3R model for detecting different types of rainfall are attributed to the complex physical mechanisms of SAR imaging, where the limited sample size constrains the monitoring of heavy rain near the eyewall, where strong winds prevail.

Minor discrepancies in rainfall locations may arise due to the time gap between NEXRAD and SAR imaging in evaluating the model's performance, rendering pixel-by-pixel comparisons challenging. Hence, this study avoided pixel-level statistical analyses for model evaluation due to inherent uncertainties that may not accurately represent the model's performance. Additionally, for TC rainfall, our emphasis is on the extent and intensity of rainfall. Therefore, spatial evaluations of rainfall inversion provide more convincing assessments of TC rainfall.

Furthermore, one limitation of our case study comparison is that it is difficult to compare quantitatively between GPM and SAR. There could be differences in the resolution, the issue of missing observation data among these two instruments, and the difference in observation time. Since hurricanes are always moving and rotating, the observation difference still leads to a possible small deviation in the location of the rain bands observed by the instruments, which cannot be completely avoided. However, we try to choose the closest GPM data with SAR observation. Therefore, we did not conduct quantitative statistical analysis but could only qualitatively compare the rainfall information provided by GPM and SAR through rainfall intensity distribution maps.

## V. Conclusion

TC rainfall typically appears as narrow and elongated rain bands, with 5–10 km diameters, that are brighter or darker than the surrounding areas in SAR images. This article develops a deep-learning-based model for estimation of TC rainfall rates from SAR images. Specifically, the developed TC3R model greatly assists in rainfall signal recognition and in rain rate inversion from C-band SAR images. This is also important because quantitative rainfall intensity estimates are valuable for the accurate inversion of TC winds and estimation of ocean surface wave fields.

Our study shows that C-band SAR is proficient in detecting medium to heavy rain bands with rates above 2.50 mm/h and below 40 mm/h. Moreover, the TC3R model excels at detecting rainfall bands around the TC eye wall. Nevertheless, its ability to detect severe rainfall in high wind speed areas near the eye wall is less pronounced.

Furthermore, we demonstrate that the rainfall patterns derived from SAR, NEXRAD, and GPM exhibit similar rainfall intensity variations and locations, verifying the results from TC3R and also suggesting potential synergy among the three observation platforms. This synergy could lead to improved monitoring and forecasting capabilities. Leveraging both NEXRAD and GPM observations may offer advantages and drawbacks, enhancing our ability to understand and predict rainfall patterns, and thus lead to improved predictions in the future. Synergistic rain observations of ground-based NEXRAD and satellite-based SAR and GPM can also potentially provide a good opportunity for monitoring the variability of TC surface rainfall structures.

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**Shanshan Mu** (Member, IEEE) received the B.S. degree in remote sensing science and technology from Shandong University of Science and Technology, Qingdao, China, in 2015, and the M.S. degree in physical oceanography from the University of Chinese Academy of Sciences, Beijing, China, in 2020.

Her research interests include the application of synthetic aperture radar (SAR) in the ocean, artificial intelligent oceanography, and tropical cyclone.



**Xiaofeng Li** (Fellow, IEEE) received the bachelor's degree in optical engineering from Zhejiang University, Hangzhou, China, in 1985, the master's degree in physical oceanography from the First Institute of Oceanography, Qingdao, China, in 1992, and the Ph.D. degree in physical oceanography from North Carolina State University, Raleigh, NC, USA, in 1997.

From 1997 to 2019, he was with the National Environmental Satellite, Data, and Information Service (NESDIS), National Ocean and Atmospheric Administration (NOAA), Washington, DC, USA, where he was involved in developing many operational satellite ocean remote sensing products. He is currently with the Institute of Oceanology, Chinese Academy of Sciences, Qingdao. His research interests include synthetic aperture radar (SAR) applications in oceanography and marine meteorology, the application of artificial intelligence in oceanographic studies, big data analytics, and satellite image processing.

Dr. Li holds editorial positions as an Associate Editor of IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING and the *International Journal of Remote Sensing*, an Editor of *Journal of Geophysical Research: Machine Learning and Computation*, and the Executive Editor-in-Chief of the *Journal of Remote Sensing*.



**Haoyu Wang** (Member, IEEE) received the B.S. degree in information and computational science and the M.S. degree in computer science and technology from Qingdao University of Science and Technology, Qingdao, China, in 2019 and 2022, respectively.

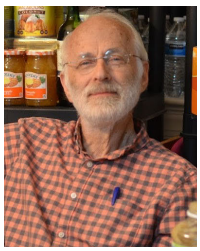
His research interests include El niño/southern oscillation forecasting and the estimation of oceanic subsurface variables using remote sensing and artificial intelligence techniques.



**Gang Zheng** (Senior Member, IEEE) received the B.Eng. degree in electronic information engineering from Zhejiang University, Hangzhou, China, in 2003, and the M.S. and Ph.D. degrees in radio physics from the University of Electronic Science and Technology of China, Chengdu, China, in 2006 and 2010, respectively.

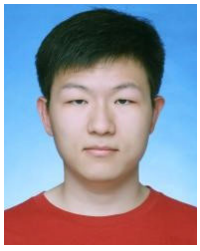
From 2010 to 2013, he was an Assistant Researcher with the State Key Laboratory of Satellite Ocean Environment Dynamics, Second Institute of Oceanography, Ministry of Natural Resources, Hangzhou, where he was an Associate Researcher from 2013 to 2020 and has been a Researcher since 2020. He has been an Adjunct Researcher with the Ocean College, Zhejiang University, and an Adjunct Professor with the College of Oceanography, Hohai University, Nanjing, China, since 2022. His research interests include ocean microwave remote sensing, artificial intelligence applications, image processing, and electromagnetic numerical modeling.

Dr. Zheng serves as an Associate Editor for *Remote Sensing in Ecology and Conservation*, an Editorial Board Member of the Ocean Section for *Remote Sensing*, and a Topic Editor for *Big Earth Data*.



**William Perrie** received the B.S. degree in physics from the University of Toronto, Toronto, ON, Canada, in 1973, and the Ph.D. degree in meteorology and oceanography from Massachusetts Institute of Technology, Cambridge, MA, USA, in 1979.

He was a Post-Doctoral Fellow in oceanography and mathematics with The University of British Columbia, Vancouver, BC, Canada, and the National Center for Atmospheric Research, Boulder, CO, USA. He is currently the Senior Scientist of the Bedford Institute of Oceanography, Dartmouth, NS, Canada, and an Adjunct Professor with Dalhousie University, Halifax, NS, Canada. His research interests include the modeling of ocean waves, air–sea fluxes, coupled atmosphere ocean interactions and impacts of climate change on these variables, and field measurements via remote sensing and in situ of winds, waves, and currents.



**Chong Wang** (Member, IEEE) received the Master of Science degree from Hohai University, Nanjing, China, in 2021, and the Ph.D. degree in marine science from the Institute of Oceanology, Chinese Academy of Sciences, Qingdao, China, in 2024.

His research interests include tropical cyclone monitoring and forecasting using remote sensing and artificial intelligence techniques.

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